

Background

As drone swarms become increasingly significant in real-world scenarios, efficient and reliable human-swarm interaction(HSI) becomes crucial. While language models offer intuitive, natural language interfaces for HSI, they can produce imprecise commands, particularly in complex environments. To address this, we introduce TruSiA, a trustworthy and situated agent-based HSI system that combines an agent-based command interpreter with situated previews and human-in-the-loop correction.

Translating the operator’s high-level instructions into executable swarm commands

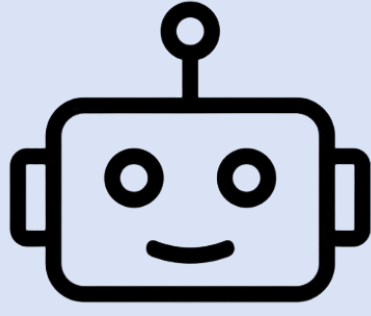
Operator’s High-Level Instruction:

Recruit three drones near the roundabout and deploy them in a loop formation above the center of the roundabout. Then, assign one drone to track a nearby green motorcycle.

Agent-based HSI Command Interpreter

You're an intelligent agent that generates structured instructions for a drone swarm system. You can issue five commands: SwarmInstantiation, TaskInstantiation, SwarmMotion, SwarmFormation, and TaskExecution. First, identify swarm and task entities with SwarmInstantiation and TaskInstantiation. Then, use SwarmMotion, SwarmFormation, and TaskExecution to assign instructions to existing swarms and tasks.

Command Interpretation Agent

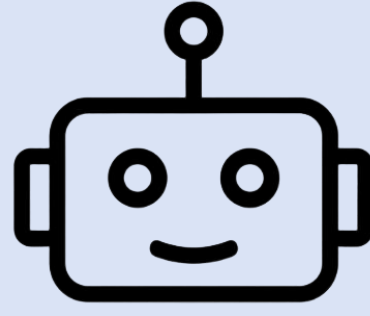


Responsible for interpreting user instructions into executable commands

Please help me identify location involved in the command.

Location Identification Result

Environment Understanding Agent



Responsible for identifying locations involved in the commands

Here is the structured output, following your prompt logic and function specifications

Instantiation Commands:

Create_Swarm_Cmd(droneNum=3, location= location near the roundabout(85, 5, 75))

Create_Task_Cmd(droneNum=1, taskType=Tracking, targetName=GreenMotorcycle(80, 0, 85))

Action Commands:

Swarm_Motion_Cmd(swarmName=Swarm 1, motionType=TakeOff)

Swarm_Formation_Cmd(swarmName=Swarm 1, formation=Loop, location=location above the roundabout(56, 30, 54))

Assign_Task_Cmd(swarmName=Swarm 1, taskName=Task 1)

Visualizing Generated Swarm Commands

Visualizing invoked commands as modules

Instantiation Commands:

Swarm 1

3 Drone From Loc1 85 5 75

Task 1

1 Drone Tracking On Green Motorcycle

Instantiation Commands:

1 Motion

Swarm 1 Take off

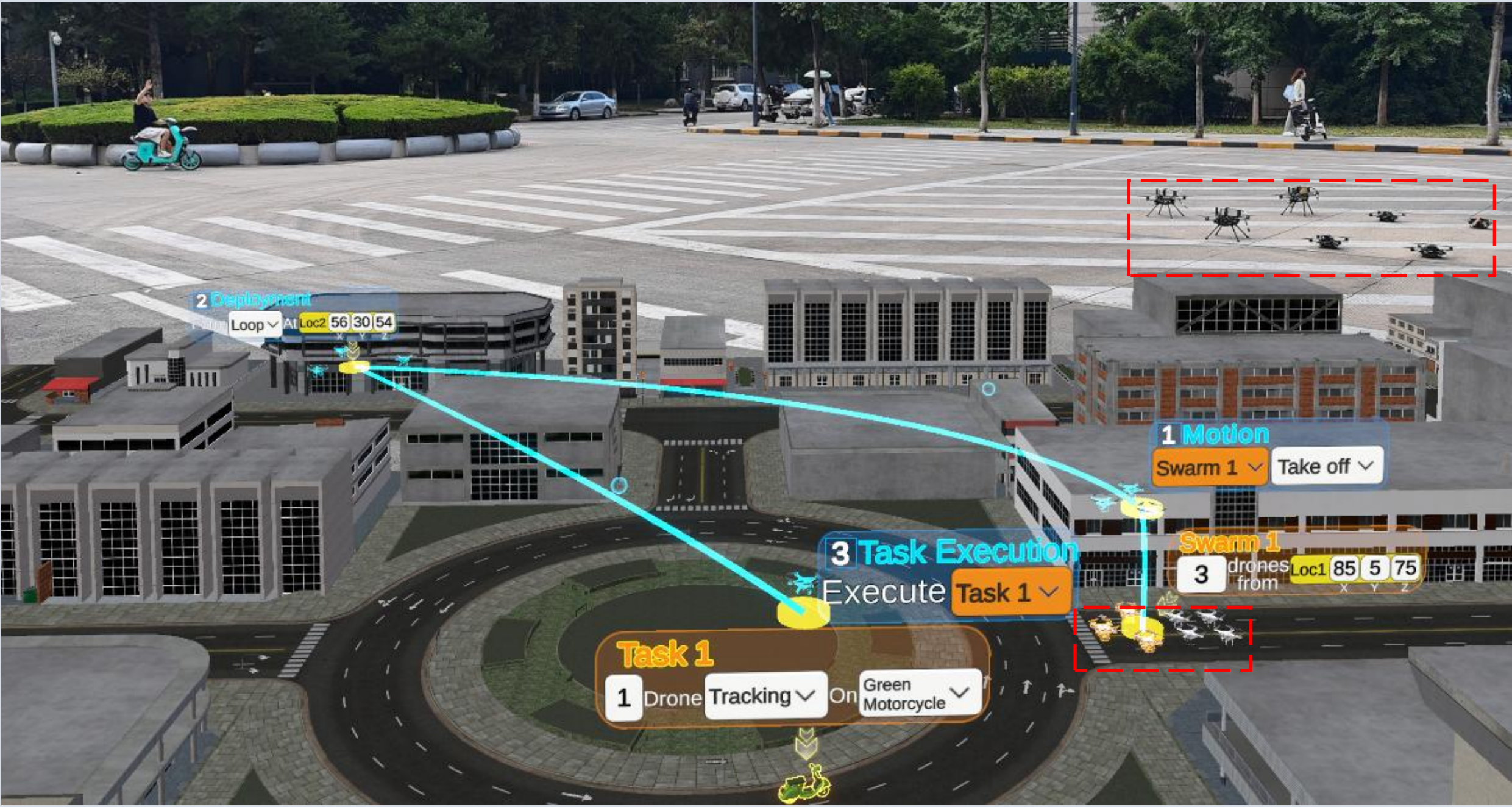
2 Formation

Form Loop At Loc2 56 30 54

3 Task Execution

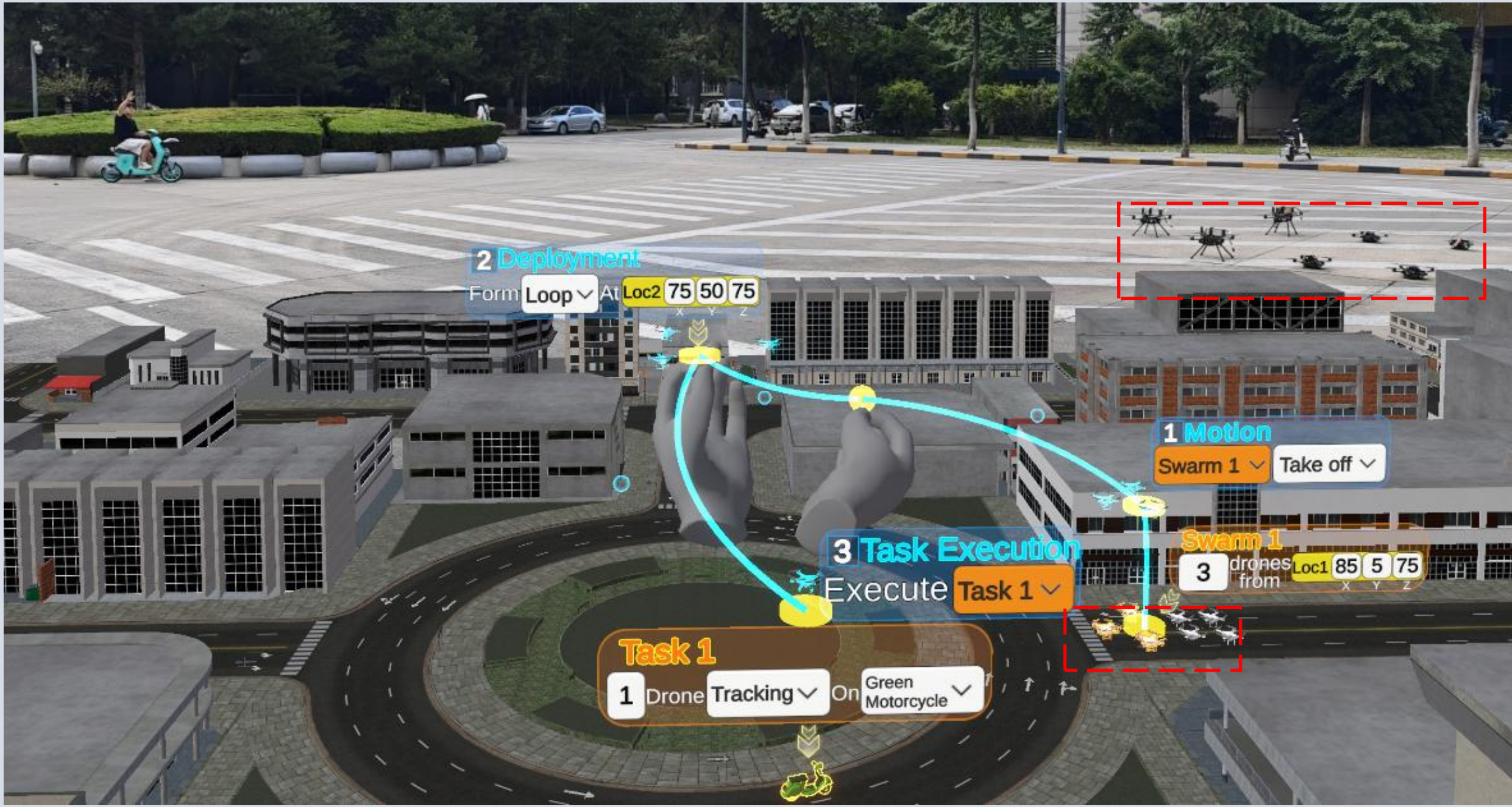
Execute Task 1

Visualizing commands on the virtual sand-table



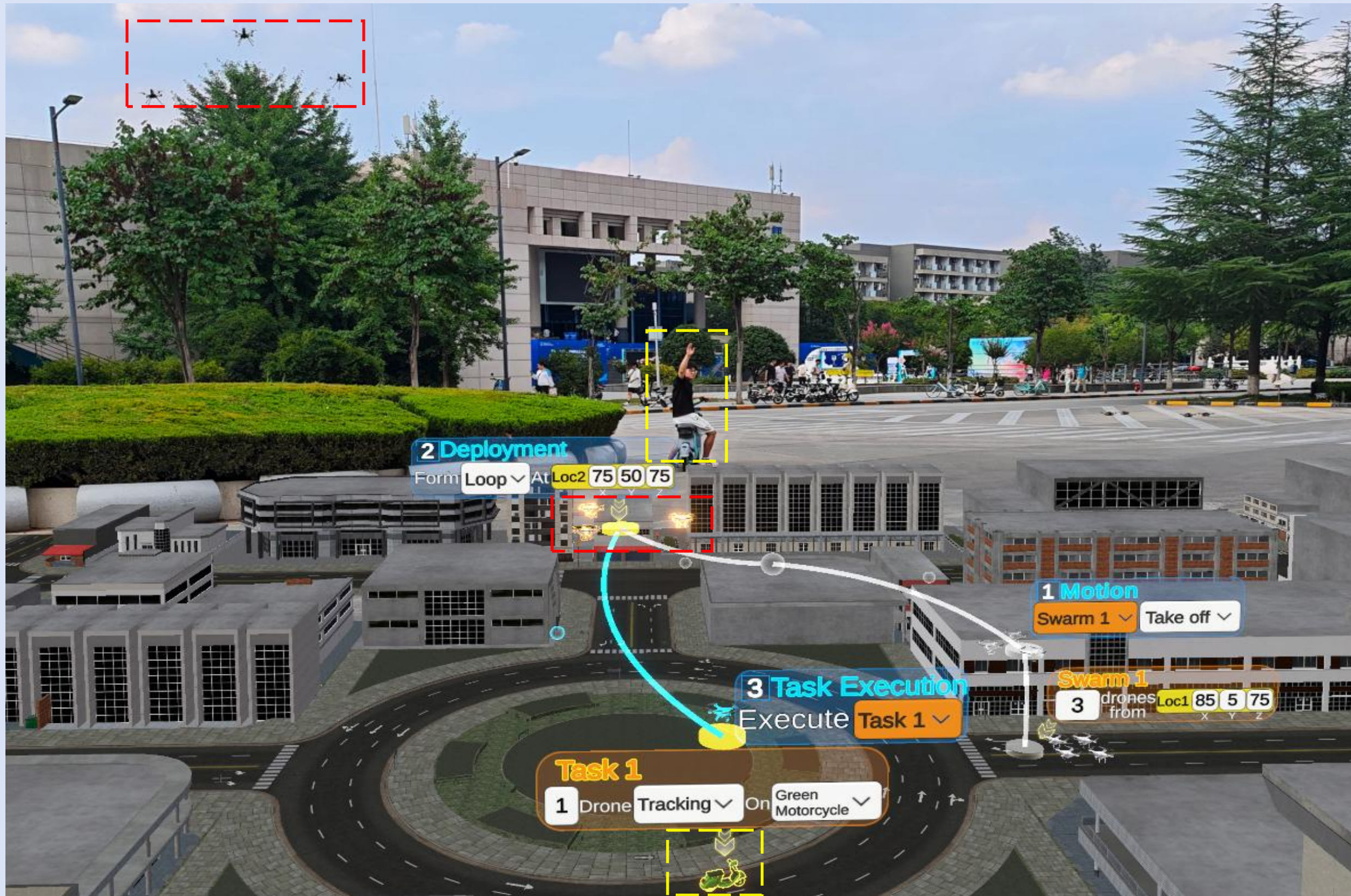
Simulating swarm trajectory based on generated commands

Not Satisfied with the result?



Adjusting visualized commands using gesture control

Execution of revised commands in physical context along with virtual mapping



Trajectory of executed commands turns into gray



Limitations and Future Work

- Managing multiple swarms simultaneously may cause visual clutter. Planning to address trajectory overlap by developing adaptive visualization and trajectory abstraction techniques.
- Conduct systematic evaluation and user studies, benchmarking against conventional swarm control software (e.g., QGC) to assess TruSiA’s scalability, usability, and efficiency.